

BigQuery data agents & Gemini Enterprise setup

Build a conversational analytics agent on BigQuery, test it, and publish it so your organisation can use it.

1. Data Agents

An agent that answers "which neighborhoods wait longest for an ambulance" in plain English, backed by SQL you can inspect, plus forecasting and anomaly detection that run as SQL functions rather than on a separate ML platform. Then the same agent inside Gemini Enterprise, available to anyone you grant access to.

2. Prerequisites

- A Google Cloud project with billing enabled
- The ability to grant IAM roles, or someone who can
- A browser signed in to the account you will build with

Enable the APIs

```
Shell
export PROJECT_ID="your-project-id"

gcloud services enable geminidataanalytics.googleapis.com --project=$PROJECT_ID

gcloud services enable cloudaicompanion.googleapis.com --project=$PROJECT_ID

gcloud services enable bigquery.googleapis.com --project=$PROJECT_ID
```

Agent permissions

Role	ID	Allows
Data Agent Creator	roles/geminidataanalytics.dataAgentCreator	Create agents. You become Owner of what you create
Data Agent Owner	roles/geminidataanalytics.dataAgentOwner	Create, edit, share, delete all agents in the project
Data Agent Editor	roles/geminidataanalytics.dataAgentEditor	View and edit all agents
Data Agent User	roles/geminidataanalytics.dataAgentUser	Chat with an agent. What most users need
Data Agent Viewer	roles/geminidataanalytics.dataAgentViewer	View configuration without using it

Data permissions

Role	ID	Why
BigQuery User	roles/bigquery.user	Run query jobs
BigQuery Data Viewer	roles/bigquery.dataViewer	Read the referenced tables
BigQuery Studio User	roles/bigquery.studioUser	Use Studio and data canvas
Gemini for Google Cloud User	roles/cloudaicompanion.user	Stateful chat conversations



Publishing permissions, needed later for section 8:

Role	ID
Discovery Engine Admin	roles/discoveryengine.admin

Creating OAuth credentials has no cleanly scoped predefined role. In practice you need Owner or Editor on the project.

Grant commands

For the person building the agent:

```
Shell
export PROJECT_ID="your-project-id"
export USER_EMAIL="builder@example.com"
for ROLE in \
roles/geminidataanalytics.dataAgentCreator \
roles/cloudaicompanion.user \
roles/bigquery.user \
roles/bigquery.studioUser \
roles/bigquery.dataViewer \
roles/discoveryengine.admin
do gcloud projects add-iam-policy-binding $PROJECT_ID \
--member="user:$USER_EMAIL" --role="$ROLE" --condition=None
done
```

For someone who only needs to **use** an agent after it exists:

```
Shell
export CONSUMER_EMAIL="analyst@example.com"
for ROLE in \
roles/geminidataanalytics.dataAgentUser \
roles/cloudaicompanion.user \
roles/bigquery.user
do gcloud projects add-iam-policy-binding $PROJECT_ID \
--member="user:$CONSUMER_EMAIL" --role="$ROLE" --condition=None
done
```

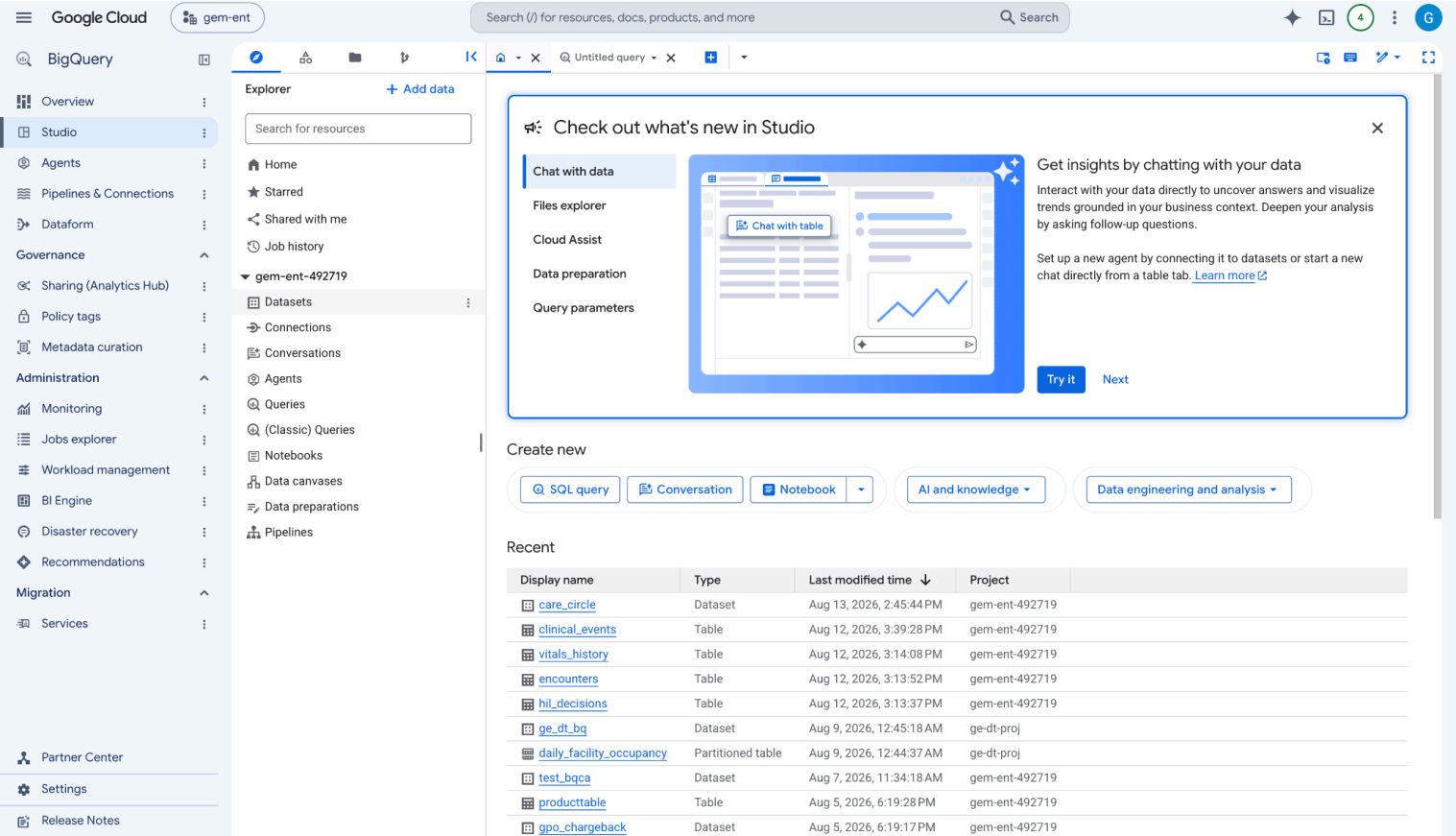
Consumers also need roles/bigquery.dataViewer on any non-public table the agent reads. For bigquery-public-data that is already handled.

Bindings take up to a minute to propagate. If a grant looks right but is not working, wait sixty seconds before debugging.



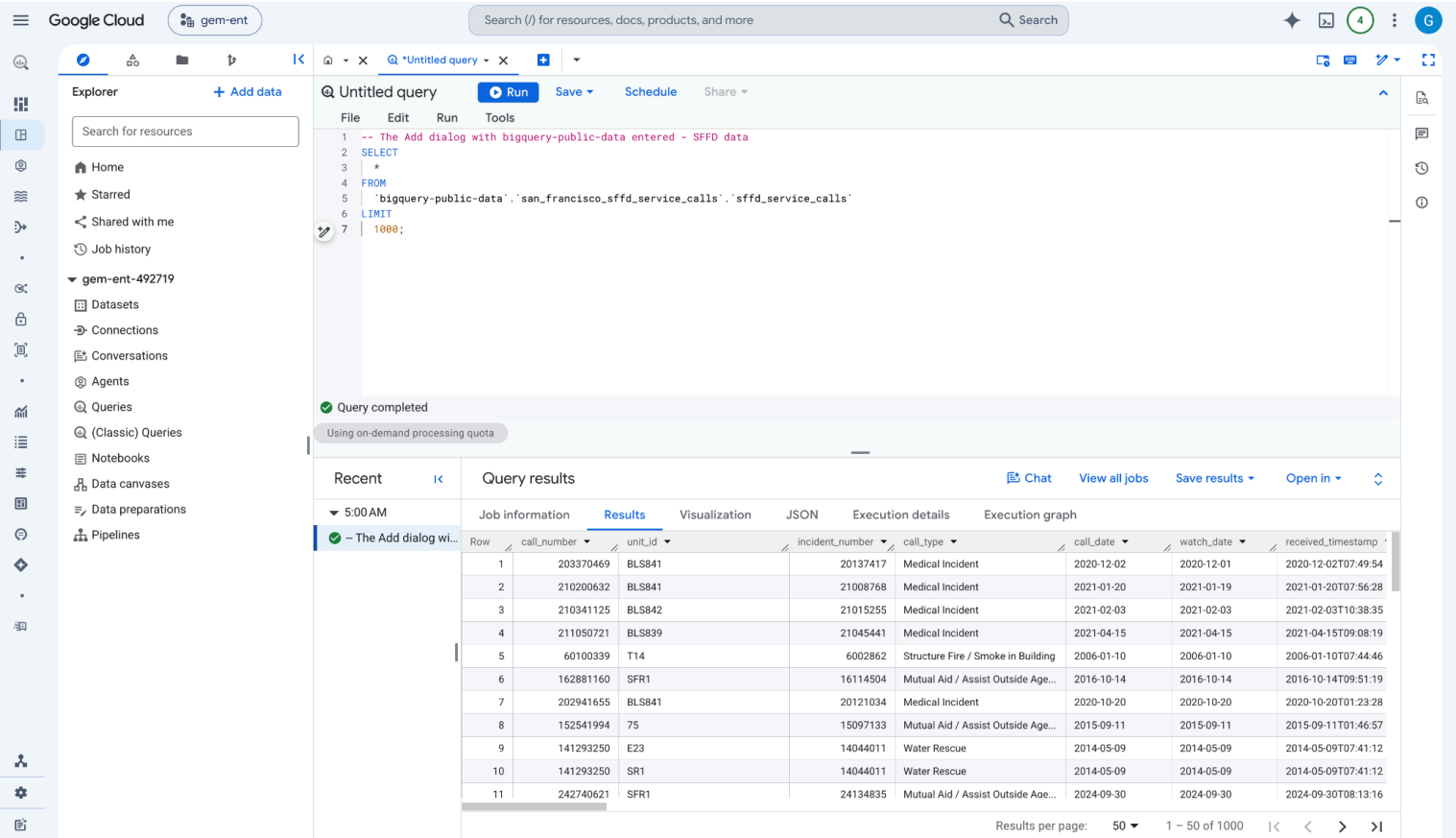
3. Open BigQuery and add the public data

Go to <https://console.cloud.google.com/bigquery> and confirm the project picker shows the project where you enabled the APIs. This catches more problems than you would expect.



[BigQuery Studio landing page with the project picker visible]

In the **Explorer** pane, click **+ Add**, choose **Star a project by name**, enter `bigquery-public-data`, and click **Star**. It stays there afterwards.



[The Add dialog with bigquery-public-data entered]

Prove you can read the data

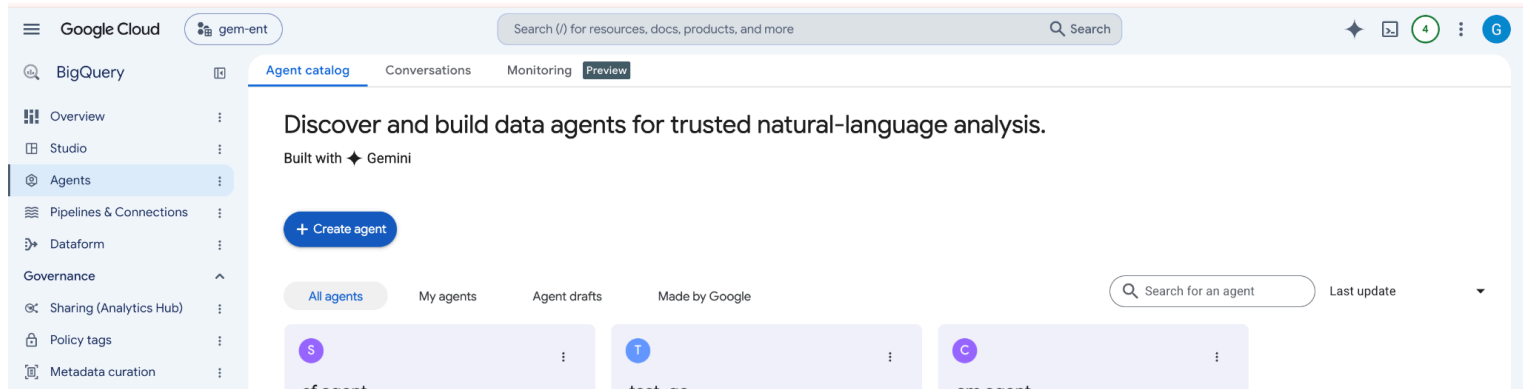
Ten seconds, and it tells you whether your permissions are right before you build anything on top of them.

```
SQL
SELECT call_date, call_type, neighborhood_name, final_priority
FROM `bigquery-public-data.san_francisco_sffd_service_calls.sffd_service_calls`
WHERE call_date >= '2026-01-01'
LIMIT 10
```

A permissions error here means `roles/bigquery.user` on your own project, not access to the public dataset. Public data is readable by anyone authenticated, but you still need permission to run a query job somewhere.

4. The configuration form, field by field

In the Explorer pane, find your own project and create a new **Data agent**. Depending on console version this is under **+ Add** or a **Data agents** node.



Agent name

Name it for the person who will use it. `SF Emergency Response Analyst` beats `sffd-agent-v2`. This string appears in Gemini Enterprise later, in a list next to other agents, to people who were not in the room when you built it.

Avoid version numbers. You will end up with three agents and no idea which is live.

Description

A sentence or two on what the agent covers and, more usefully, what it does not. Users read this to decide whether to bother asking.

Good: "Answers questions about San Francisco emergency dispatch, including call volume, response times, and neighborhood patterns. Covers 2000 to present. Does not include patient outcomes or hospital data."

Bad: "AI agent for data analysis."



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gem-ent

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BigQuery

Overview

Studio

Agents

Pipelines & Connections

Dataform

Governance

Sharing (Analytics Hub)

Policy tags

Metadata curation

Administration

Monitoring

Jobs explorer

Workload management

BI Engine

Disaster recovery

Recommendations

Migration

Services

Partner Center

Settings

Release Notes

Agent catalog

Conversations

Monitoring

Preview

← sffd

Changes not saved

Save

Publish

Share

Create conversation

bigquery-public-data.san_francisco_sffd_service_calls.sffd_se...

Customize

Add source

Instructions ⓘ

Show examples

give rich visual insights always

Verified queries (0)

Add trusted questions and answer pairs as reference for the agent to improve response accuracy. Select from generated suggestions or create your own. [Learn more](#)

Add query

View Gemini-generated suggestions

Glossary (0)

View glossary terms imported from Knowledge Catalog and create new glossary terms to help your agent interpret user prompts. [Learn more](#)

Add term

Agent settings

Model

Choose between Generally Available (GA) or a mix of GA and Preview releases. Your selection determines which models are available to users in the conversation. Available models are periodically upgraded to the latest versions.

Allow both Preview and Generally Available model(s)

Preview

S sffd

sffd data insight agent

Ask a question

Fast

See best practices for better results. The responses from Conversational Analytics in BigQuery may not be complete or accurate. [Learn more](#)

Knowledge sources

The tables the agent can read.

Start with one table. Agents get *less* reliable as you attach more, not more capable. Every extra table widens the space of queries the agent chooses between, and unrelated tables give it opportunities to invent joins that produce plausible nonsense.

If you attach several, they should be tables a human analyst would genuinely join.

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Overview

Studio

Agents

Pipelines & Connections

Dataform

Governance

Sharing (Analytics Hub)

Policy tags

Metadata curation

Administration

Monitoring

Jobs explorer

Workload management

BI Engine

Disaster recovery

Recommendations

Migration

Agent catalog

Conversations

Monitoring

Preview

← Add knowledge source

Once this agent is saved, its region cannot be changed.

The storage region for this agent and its conversations is automatically set based on where your selected knowledge sources are stored. You can override this setting, but cannot change it after the agent is saved. [Learn more](#)

Region: US

Sources

Graphs

Preview

Search for sources

sffd

Search results

Filter

Filter tables

Table name

Description

Project

Last viewed

Location

☒

sffd_service_calls

Dataset: san_francisco_sffd_service_calls

bigquery-public-data

August 14, 2026

us

☐

sffd_service_calls

Dataset: san_francisco

bigquery-public-data

January 25, 2018

us

Selected sources

1 / 100

bigquery-public-data.san_francisco_sffd_...

Instructions and context

The most important field on the form, and the one people leave blank.

The agent reads this before writing any SQL. Without it the agent still works, but it makes assumptions about your columns that are hard to spot in the output. The classic failure is a priority or severity column: the agent cannot know whether 1 means urgent or trivial, so it guesses, and half your answers come back backwards.

Write it in plain English. You are briefing a competent new analyst on their first day. Cover four things:



- 1. **What one row means.** A transaction, an event, a daily snapshot, one unit dispatched?
- 2. **Any column whose meaning is not obvious from its name.** Codes, flags, direction of scales.
- 3. **How to derive the metrics people actually ask about.** If response time is the gap between two timestamps, say so.
- 4. **Known traps.** Negative values, partial current day, duplicate grain.

Verified prompts

Question and query pairs you have checked yourself. When a user asks something similar, the agent leans on your version rather than improvising.

Three to five well-chosen ones do more good than twenty vague ones. Good candidates: questions your team asks weekly, anything with a non-obvious definition such as what counts as an active customer, and anything the agent has already got wrong once.

Agent catalogConversationsMonitoringPreview

← Review suggested verified queries

ⓘ

Check suggestions for accuracy. You can edit the question or SQL before adding. [Learn more](#)

✓

✳️ What are the top 10 most common service call types in San Francisco? ▾

sffd_service_calls

✓

✳️ What is the average time taken for units to arrive on scene for different call type groups? ▾

sffd_service_calls

✓

✳️ Which neighborhoods had the highest number of emergency incidents (Code 3) in the last year, and what was their average on-scene duration? ▾

sffd_service_calls

✓

✳️ What are the peak hours of the day when service calls are most frequently received, and how many calls are received during those hours? ▾

sffd_service_calls

✓

✳️ What percentage of units dispatched were Advanced Life Support (ALS) units each month in 2016? ▾

sffd_service_calls

Add

Cancel

Agent catalog

Conversations

Monitoring

Preview

← sffd

Changes not saved

Save

Publish

Share

Create conversation

Knowledge sources

Add tables, UDFs, and other resources. Ensure agent users have access to these resources.

bigquery-public-data.san_francisco_sffd...

Customize

Add source

Instructions ⓘ

Show examples

give rich visual insights always

Verified queries (5)

Add trusted questions and answer pairs as reference for the agent to improve response accuracy. Select from generated suggestions or create your own. [Learn more](#)

What are the top 10 most common service call types in San F...

What is the average time taken for units to arrive on scene fo...

Which neighborhoods had the highest number of emergenc...

Show more

Add query

View Gemini-generated suggestions

Glossary (0)

View glossary terms imported from Knowledge Catalog and create new glossary terms to help your agent interpret user prompts. [Learn more](#)

Preview

Save agent changes

hi

Show thinking

Medical Incidents are by far the most common service call type in San Francisco, accounting for over 2.5 million incidents and nearly 4.9 million total calls, as shown in the top 10 most common service call types analysis.

Here's the query result for Most Common Service Calls.

	call_type	total_incidents	total_calls
5	CITIZEN ASSIST / Service Call	94,793	117,055
6	Other	92,100	128,019
7	Outside Fire	76,580	103,324

Top 10 Most Common Service Call Types in San Francisco

Chart

Table

Ask a question

Fast

See best practices for better results. The responses from Conversational Analytics in BigQuery may not be complete or accurate. [Learn more](#)

Glossary

Business terms mapped to what they mean in the data. This is where you settle arguments once instead of every quarter. If "active store" means one with a transaction in the last 90 days, put it here and every answer uses that definition. Without it the agent picks something reasonable, and reasonable is not the same as your company's definition.

Labels

Standard key-value labels. They do not change behaviour, they exist for organising and reporting.

None

env = dev | test | prod

owner = team-name

domain = ops | retail | finance

data-source = public | internal

Set them now. Retrofitting labels across thirty agents is nobody's idea of a good afternoon.

Maximum billed bytes

A ceiling on how much data a single agent query may scan. Anything exceeding it fails instead of running. This is your cost guardrail, and you want it set before you hand the agent to anyone: a user typing an innocent-sounding question can generate a full table scan without knowing it.

Situation	Suggested limit
Small curated table, under 1 GB	1 GB
Single table up to a few GB	5 GB



Situation	Suggested limit
Large table with wide columns	10 GB
No idea yet	5 GB, then watch what fails

Set it low first. Raising a limit after a complaint is a two-minute fix. Explaining a surprise bill is not.

Forecasting, anomaly detection and driver analysis run as SQL functions. No separate ML platform, no model training, no feature store.

Publishing Data Agent as A2A agent to Gemini Enterprise :

Create the OAuth client

Go to **APIs & Services**, then **Credentials**.

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Search

Google Auth Platform / Clients / Client: 859413997574-ufmo1d5j8cck8u9h2oeng2b595ph1mui.apps.googleusercontent.com

Google Auth Platf...

Overview

Branding

Audience

Clients

Data Access

Verification Center

Settings

Client ID for Web application

Delete

Name *

Web client abc

The name of your OAuth 2.0 client. This name is only used to identify the client in the console and will not be shown to end users.

The domains of the URIs you add below will be automatically added to your [OAuth consent screen](#) as [authorized domains](#).

Authorized JavaScript origins

For use with requests from a browser

+ Add URI

Authorized redirect URIs

For use with requests from a web server

URIs 1 *

https://vertexaisearch.cloud.google.com/static/oauth/oauth.html

URIs 2 *

https://vertexaisearch.cloud.google.com/oauth-redirect

+ Add URI

Note: It may take 5 minutes to a few hours for settings to take effect

Save

Cancel

Additional information

Client ID

859413997574-ufmo1d5j8cck8u9h2oeng2b595ph1mui.apps.googleusercontent.com

Creation date

August 12, 2026, 4:46:57 PM GMT-4

Last used date

August 13, 2026 (Note: this data could be delayed by a day or more.)

Inactive OAuth clients are subject to deletion if they are not used for 6 months. You will be notified of deletion due to inactivity, and can restore clients up to 30 days after deletion. [Learn more](#)

Client secrets

If you are in the process of changing client secrets, you can manually rotate them without downtime. [Learn more](#)

Viewing and downloading client secrets is no longer available. If you have lost the secret below, please add a new one. Remember to store client secrets securely and treat them with extreme care. Secrets should never be checked into code repositories. [Learn more](#)

Client secret

****QItg

Creation date

August 12, 2026, 4:46:57 PM GMT-4

Status

Enabled

+ Add secret

Confirm the project selector shows the project containing your BigQuery dataset. This is the most common mistake in the whole process.

Click **Create credentials**, then **OAuth client ID**. Application type **Web application**. Name it something you will recognise, such as gemini-enterprise-bq-agent.

Under **Authorized redirect URIs**, add **both** of these. Both are required and they must match exactly.

None

https://vertexaisearch.cloud.google.com/oauth-redirect

None

https://vertexaisearch.cloud.google.com/static/oauth/oauth.html

A mismatch here is the single most common failure in this process, and the error you get later does not tell you that is the problem. Copy and paste rather than typing, and check for a trailing slash.

Click **Create**, then **Download JSON**. That file has the Client ID, Authorization URI, Token URI and Client secret. You need all four. Treat it as a secret, because it is one.

If your console shows **Google Auth Platform** rather than the older OAuth consent screen menu, that is expected. Settings are split across **Branding**, **Audience** and **Clients**, and your client is under Clients.

Get the agent card from BigQuery

Back in BigQuery, click the **three dots** next to your agent, then **Edit**, and copy the JSON for publishing to Gemini Enterprise.

Paste it into a scratch file. You will be switching tabs, and clipboards lose things at the worst moment.

Agent catalog

Conversations

Monitoring

Preview

← sffd

Knowledge sources

bigquery-public-data.san_francisco_sffd...

Add source

Instructions

Give rich visual insights always

Verified queries (5)

What are the top 10 most common service call types in San F...

What is the average time taken for units to arrive on scene fo...

Which neighborhoods had the highest number of emergenc...

Glossary (0)

Preview

hi

Show thinking

Medical Incidents are by far the mo

incidents and nearly 4.9 million tota

Here's the query result for Most Co

	call_type
5	CITIZEN ASSIST / SERVICE CALL
6	Other
7	Outside Fire

Top 10 Most Common Service Call Types

Call Type	Count
Medical Incident	4,900,000
Alarms	1,200,000
Structure Fire / Smoke in Building	1,100,000
Traffic Collision	800,000
Citizen Assist / Service Call	700,000
Other	600,000
Outside Fire	500,000
Electrical Hazard	400,000
Elevator / Escalator Rescue	300,000
Gas Leak (Natural and LP Gases)	200,000

Ask a question

Publishing channels

Updates available to publish - Draft

Your agent will appear in these channels for you and anyone you share it with.

BigQuery

Conversational Analytics API

Data Studio

Additional channels

Integrate via A2A (Agent2Agent)

Copy JSON

Publish agent

Cancel

Register the agent

Go to <https://console.cloud.google.com/gemini-enterprise> and create an app if you do not have one.

Click **Add Agent**, choose **Custom agent via A2A**, and paste the JSON.



Google Cloud

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Gemini Enterprise

Overview

Connected data stores

Actions

Prompt chips

Configurations

Agents

Skills

Security

Integration

Analytics

Gemini Notebook

CodeAssist

Manage subscriptions

Apps

gemini-enterprise-1

Agents

Agents

All

Google-made

Our agents

Add agent

By default, observability and user content logging is disabled for all supported agents. To change

Agents table

Filter

Enter property name or value

Display name	Agent ID	Agent type	Agent s
Core Assistant	core_assistant	-	-
My Agent	9178434196552783659	Employee-made	Priv
My Agent	17202439832173241740	Employee-made	Priv
My Agent	12153952562897010600	Employee-made	Priv
Onboarding Guide	6520446294155630873	Employee-made	Priv
My Agent	4234608747252933256	Employee-made	Priv
Smart Care Triage	7631615780201079867	Agent Engine	Enab
Deep Research	deep_research	Google-made (Deep Research)	Enab

Add an agent

Choose an agent type

Please choose the type of agent you would like to add

Custom agent via Agent Runtime

Agents built by your organization that follow your company policies

Add

Custom agent via Dialogflow

Agents built using the Dialogflow framework

Add

Custom agent via A2A

Custom Agents deployed anywhere that speak A2A protocol

Add

Agents via Marketplace

Agents already procured or obtained via the Marketplace

Add

Agents from Agent Registry

Agents in a registry within the Agent Gateway configured for this Gemini Enterprise instance

Add

Google Cloud

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Gemini Enterprise

Overview

Connected data stores

Actions

Prompt chips

Configurations

Agents

Skills

Security

Integration

Analytics

Gemini Notebook

CodeAssist

Manage subscriptions

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Google-made

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Display name	Agent ID	Agent
Core Assistant	core_assistant	-
My Agent	9178434196552783659	Emplc
My Agent	17202439832173241740	Emplc
My Agent	12153952562897010600	Emplc
Onboarding Guide	6520446294155630873	Emplc
My Agent	4234608747252933256	Emplc
Smart Care Triage	7631615780201079867	Agent
Deep Research	deep_research	Googl

Add agent

1 Import agent

2 Agent authorization

Import Agent

Import agent details by pasting JSON content.

Agent Card JSON

mcilGhlaWdodD0iMTAwliB2aWV3Qm94PSiwiDAgMjQgMjQilHdpZHRoPSixMDAiPjxwYXRolGQ9Ik02LjczLDEwLjgzdjljuNjNBNC45MSw0LjksLDAsMCwwLDguNDQsMTUuMIYxMC44M1oiLz48cGF0aCBkPSJNOS44OSw4LjQxdjcuNTNBNy42Miw3LjYyLDAsMCwwLDExLDE2LDgsOCwwLDAsMCwwMiwxNiY4LjQxWiivPjxwYXRolGQ9Ik0xMy42NCwwMS44NnYzLjI5YTUsNSwwLDAsMCwwLjctMS44MlYxMS44Ni0iLz48cGF0aCBkPSJNMTcuNzQsMTYuMzJsLTUuNDIsMS40MmEuNDluNDIsMCwwLDAsMCwwNmwwZLjU0LDMuNTRhLjQyLjQyLDAsMCwwLC41OSwwbDEuNDMtMS40M2EuNDluNDIsMCwwLDAsMC0uNTIsLTMuNTQtMy41NGEuNDluNDIsMCwwLDAtLjYsMClvPjxwYXRolGQ9Ik0xMSwyYTksOSwwLDEsMCw5LDksOSw5LDAsMCwwLTktOW0wLDE1LjY5QTYuNjgsNi42OCwwLDEsMSwwNy42OSwwMSw2LjY4LDYuNjgsMCwwLDEsMTesMTcuNjkiLz48L3N2Zz4="

Preview Agent Details

Agent details

Icon

Agent name

sffd

Agent description

sffd data insight agent

Agent URL

https://geminidataanalytics.us.rep.googleapis.com/v1beta/a2a/projects/gem-ent-492719/d799dce54426

Protocol version

1.0

Agent capabilities

Streaming

Gemini Enterprise A2UI Thoughts extension

Ability to render

Ability to render SDUI charts

Google Data Analytics Client ID extension

Google Gemini Enterprise authorization extension

Google Gemini Enterprise

Server-Sent Events (SSE) extension. Indicates that the agent supports SSE, promp

GDA Stateless extension

BQ Dataset Information

Agent skills

Provides data analysis assistance

Enter the authorization details from your downloaded JSON: Client ID, Client Secret, Authorization URI, Token URI, Scopes. **Skip Tool authorization**, it is not needed for a BigQuery data agent.

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Gemini Enterprise

Overview

Connected data stores

Actions

Prompt chips

Configurations

Agents

Skills

Security

Integration

Analytics

Gemini Notebook

CodeAssist

Manage subscriptions

Apps > gemini-enterprise-1 > Agents

Agents

All Google-made Our agents

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My Agent	17202439832173241740	Employee
My Agent	12153952562897010600	Employee
Onboarding Guide	6520446294155630873	Employee
My Agent	4234608747252933256	Employee
Smart Care Triage	7631615780201079867	Agent
Deep Research	deep_research	Google

Add agent

Import agent

Agent authorization

Agent authorization

Create an authorization for your agent

Authenticate agent

Provide the following details from your OAuth provider. Some fields may be pre-filled from the agent card. No need for auth if the agent is publicly available or there is already permission for runtime.

Provider credentials

Client ID

Client secret

Additional details

These values are typically populated from the agent card. You can override them, but exercise caution as incorrect details can break the integration.

Authorization URL

Token URL

Scopes

PKCE verification enabled

Finish

Skip & Finish

Cancel

Under **Agent authorization**, the block looks like this:

```
JSON
{
  "client_id": "<client_id>",
  "client_secret": "<client_secret>",
  "authorization_uri":
  "https://accounts.google.com/o/oauth2/v2/auth?client_id=<client_id>&redirect_uri=
  https%3A%2F%2Fvertexaisearch.cloud.google.com%2Fstatic%2Foauth%2Foauth.html&scope=h
  ttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform&include_granted_scopes=true
  &response_type=code&access_type=offline&prompt=consent",
  "token_uri": "https://oauth2.googleapis.com/token",
  "scope": "https://www.googleapis.com/auth/cloud-platform"
}
```

Three things to get right:

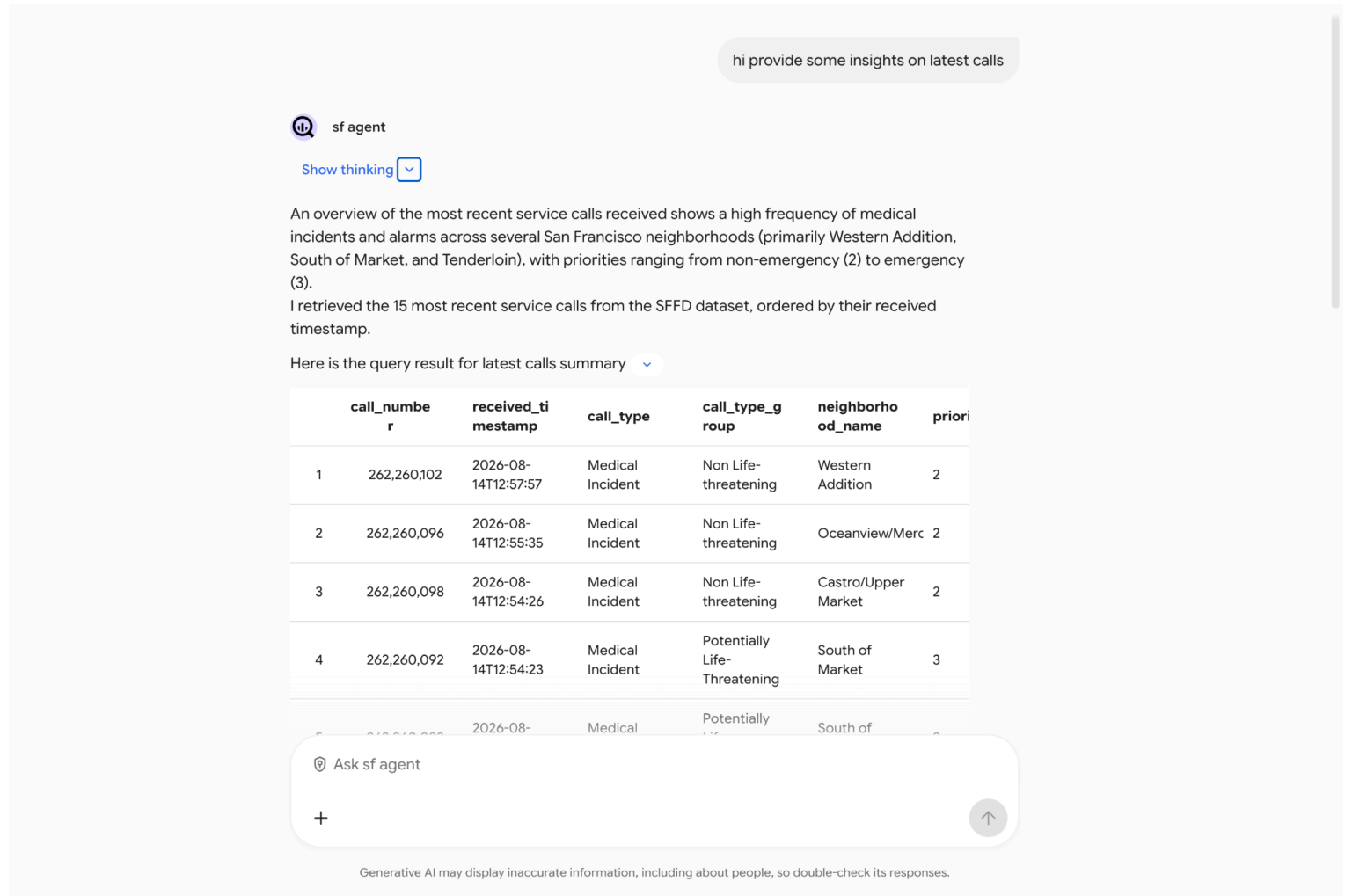
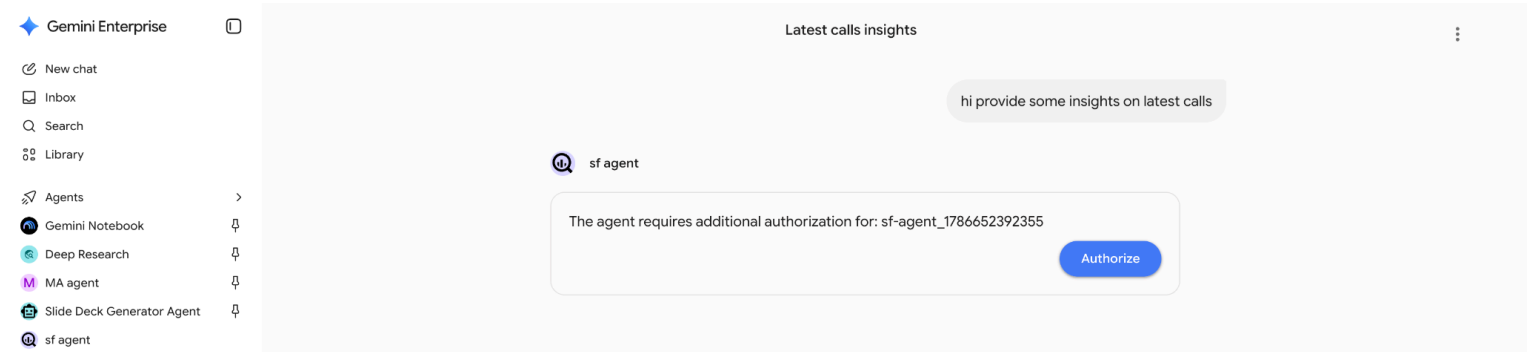
- `<client_id>` appears **twice**. Once as the `client_id` field, and again inside the query string of `authorization_uri`. Replace both.
- The `redirect_uri` inside `authorization_uri` is URL-encoded. Leave the encoding exactly as it is. Decoded it is the second URI you registered above.
- `access_type=offline` and `prompt=consent` are what produce a refresh token. Remove them and users get re-prompted constantly.

Connect the agent.

Use it, then prove it works for someone else

The agent now appears under **Agents** in your app. Ask it a question you already tested in BigQuery and you should get the same answer, rendered as a chart.

The first time any user opens the agent they get an OAuth consent prompt. That is the system working. Tell your users to expect it, because an unexpected consent screen makes people close the tab.



Then sign in as a different user, ideally one with no BigQuery console access, and ask the same question. They should get the same answer without ever seeing BigQuery. That is what publishing bought you.

Troubleshooting

Setup and permissions

Symptom	Cause and fix
No data agent option in the console	<code>geminidataanalytics.googleapis.com</code> not enabled, or enabled under a minute ago. Enable and reload
Permission denied creating an agent	Missing roles/ <code>geminidataanalytics.dataAgentCreator</code> , on the project not a dataset
Test query fails on permissions	Missing roles/ <code>bigquery.user</code> . Running a job is separate from reading public data
Chat fails but the agent saves fine	Missing roles/ <code>cloudaicompanion.user</code> . Easy to miss because creation works without it



Symptom	Cause and fix
Works for you, not for a colleague	They need dataAgentUser plus bigquery .user. Agent access and data access are separate
Roles granted, console still denies	Organisation-level deny policy or Principal Access Boundary. If you hold Owner and are blocked, it is almost always this

Agent quality

Symptom	Cause and fix
Confidently wrong about priority or severity	Instructions do not state the direction of the scale
Totals too high or too low	Grain mismatch, usually a header table joined to line items. State the grain in the instructions
Agent picks the wrong table	Too many knowledge sources. Remove the unrelated ones
Queries slow or expensive	Lower maximum billed bytes and add date guidance to the instructions

AI functions

Symptom	Cause and fix
AI . KEY_DRIVERS errors on non-negative values	Metric column contains negatives. Filter them in the subquery
AI . DETECT_ANOMALIES "Unrecognized name"	The function renames its output. Use time_series_timestamp and time_series_data
AI . DETECT_ANOMALIES errors before running	Pass exactly one of cutoff_timestamp or last_n_points. Neither fails
Forecast returns a flat line	Series too sparse or too short. It needs dense, regularly spaced points
Everything recent flagged as an anomaly	Strong growth trend in the series. Use a stable series instead
Latest day looks like a collapse	You included today, which is partial. Filter to yesterday

Publishing

Symptom	Cause and fix
redirect_uri_mismatch on consent	URI in authorization_uri does not match the registered client. Check URL encoding and trailing slashes
Consent appears then fails silently	Usually only one of the two redirect URIs was registered. Both are required
Agent registers but returns no data	User has agent access but not data access. Grant bigquery .user and bigquery .dataViewer
Access denied on the Credentials page	Owner or Editor missing, or an org deny policy. Check IAM & Admin > Deny
Users re-prompted for consent constantly	access_type=offline or prompt=consent missing, so no refresh token is issued
OAuth client in the wrong project	It must be the project containing the dataset. Create a new client in the right one
Agent answers differently than in BigQuery	You copied a stale agent card. Editing the agent after publishing means republishing